Phase-Specific Effects of AI in Self-Regulated Learning: A Meta-Analysis of a Decade of Interventions

Abstract: *This meta-analysis addresses inconsistent findings on AI's role in self-regulated learning (SRL) by systematically examining 35 empirical studies (2013-2025) through Zimmerman's cyclical model. Results reveal a moderate overall effect of AI interventions (g=0.507), with significantly stronger impacts during the task performance phase (g=0.574). A key finding is the superior efficacy of generative AI (g=0.799) over rule-based and data-driven paradigms across all SRL phases. Critically, this synthesis highlights a "performance-competence divide," where AI excels at scaffolding immediate tasks but its ability to cultivate transferable competence remains unproven. This research offers a nuanced, evidence-based map for designing future AI systems that foster genuine learner autonomy.*

Keywords: Self-regulated learning, Artificial intelligence, large language models, Educational technology, Meta-analysis

Research Objective

While artificial intelligence (AI) shows considerable promise for supporting self-regulated learning (SRL), the empirical evidence remains fragmented, with inconsistent findings across different learning phases and a lack of synthesis that accounts for AI's rapid technological evolution. This meta-analysis addresses this critical gap by systematically synthesizing a decade of intervention research (2013-2025). Guided by Zimmerman's (2002) cyclical model, the study's primary objective is to quantify the overall and phase-specific effects of AI support on SRL. It further aims to disaggregate these effects by examining how they vary across three distinct technological paradigms—rule-based, data-driven, and generative AI—and to identify key moderators that shape their effectiveness. The overarching goal is to provide a comprehensive, evidence-based map of AI's evolving role in SRL, moving beyond a monolithic view to offer nuanced insights for designing technologies that foster genuine learner autonomy.

Theoretical Framework

A Cyclical Model of Self-Regulated Learning

This study is grounded in the social cognitive view of self-regulated learning (SRL), which posits that learners proactively manage their cognition, motivation, and behavior to achieve academic goals (Zimmerman, 1989). While SRL theory has evolved to encompass various dynamic perspectives (e.g., Bandura, 1991; Winne & Hadwin, 1998) , this meta-analysis adopts Zimmerman's (2002) three-phase cyclical model as its primary analytical lens. This model delineates SRL into three recursive phases: a **forethought** phase involving pre-task analysis and goal setting; a **performance** phase characterized by in-task strategy use and self-monitoring; and a **self-reflection** phase for post-task evaluation and attribution. Its widespread use as an operational framework in technology-enhanced learning research (Wong et al., 2019) provides a robust, process-oriented structure to systematically map and evaluate the effects of diverse AI interventions on the core components of individual learning.

Synthesizing AI's Evolving Role in the SRL Cycle

While prior research affirms the general benefits of technology for supporting learner autonomy (Dent & Koenka, 2016; Dignath & Büttner, 2018), the rapid growth of the SRL field itself (see Figure 1) has outpaced systematic syntheses of AI's specific role. Existing meta-analyses have often concentrated on particular technologies or treated SRL as a unified construct, thereby obscuring how different AI paradigms support the distinct cognitive and metacognitive demands of each regulatory phase (Chen et al., 2024; Lu & Lin, 2024). This study addresses this critical gap by moving beyond a monolithic view of AI. It synthesizes evidence through the dual lens of Zimmerman's (2002) model and the technological evolution of AI (see Figure 2)—from rule-based systems to data-driven platforms and generative AI. This approach allows for a nuanced examination of how the alignment between specific AI affordances and the psychological processes within each SRL phase contributes to intervention effectiveness, providing a more detailed map for future research and evidence-based practice.

Method

This study followed the meta-analytic procedures outlined by Pigott and Polanin (2020) and adhered to PRISMA reporting guidelines (Chen et al., 2024).

Data Sources and Inclusion Criteria

A systematic search for empirical studies published between 2013 and 2025 was conducted across four major databases: ERIC, Ei Compendex, Scopus, and Web of Science. The search and screening process, detailed in the flow diagram (see Figure 3), yielded a final sample of 35 studies. To be included, studies had to: (1) examine an AI-supported SRL intervention; (2) use an experimental or quasi-experimental design comparing an intervention to a control group; (3) provide measures of SRL processes or direct outcomes; (4) report sufficient data for effect size calculation; and (5) achieve a Medical Education Research Study Quality Instrument (MERSQI) score of 10 or higher (Sawatsky et al., 2015). The final robust sample (mean MERSQI = 13.07), comprising 31 journal articles and 4 conference proceedings, yielded 133 independent effect sizes for analysis.

Coding and Data Analysis

A systematic coding scheme was developed to extract study characteristics and classify AI paradigms (see Table 1). To enable a phase-specific analysis, SRL outcome variables from each study were mapped onto Zimmerman's (2002) forethought, performance, or self-reflection phases (see Table 2). The coding process demonstrated strong interrater reliability (Cohen's κ = .85-.88) (Landis & Koch, 1977). Due to significant heterogeneity across all subgroups ($I^2 > 75\%$), a random-effects model was employed for all analyses (Higgins & Thompson, 2002). Effect sizes were calculated as Hedges' g to account for small sample sizes, and all analyses were conducted using Stata 18 with robust variance estimation (Hedges & Tipton, 2010).

Results

The analysis of 35 primary studies yielded 133 effect sizes. Initial heterogeneity tests confirmed substantial variance across studies (overall $I^2 = 89.81\%$, $p < .001$), justifying the use of a random-effects model for all subsequent analyses.

The Evolution of AI Paradigms for SRL Support

In response to RQ1, our chronological analysis of the included studies identified three co-existing technological paradigms for AI-supported SRL, the distribution of which is illustrated in Figure 4. The initial paradigm consists of rule-based systems, which rely on predefined expert knowledge to provide structured but relatively rigid scaffolding (e.g., Duffy & Azevedo, 2015). This was followed by the emergence of data-driven personalization systems, characterized by their capacity to analyze real-time learner data to dynamically tailor feedback and learning paths (e.g., Harati et al., 2021). The most recent paradigm features generative AI ecosystems, which leverage large language models (LLMs) to act as proactive, dialogic partners in the learning process (e.g., Ng et al., 2024).

Overall and Phase-Specific Effects of AI Interventions

As detailed in Table 3, AI-supported interventions demonstrated a moderate and statistically significant overall effect on SRL ($g = 0.507$, 95% CI [0.384, 0.630]). The impact of AI varied across the SRL cycle, with the strongest effect observed during the performance phase ($g = 0.574$). The effects were smaller for the self-reflection ($g = 0.464$)

and forethought phases (g = 0.401). Sensitivity analyses confirmed the stability of these estimates. However, visual inspection of the funnel plots (see Figure 5 and Figure 6) and Egger's regression tests indicated the potential for publication bias, particularly for the overall sample and the performance phase (see Table 4), suggesting results should be interpreted as potential upper boundaries.

Subgroup and Moderator Analyses

Subgroup analyses by AI paradigm, which reflect a clear technological progression over the past decade, revealed significant differences in effectiveness (see Table 5). Generative AI interventions yielded a large overall effect (g = 0.799), which was substantially greater than those of rule-based (g = 0.534) and data-driven systems (g = 0.318). This advantage for generative AI was consistent across all SRL phases, with a particularly strong effect on performance (g = 0.938). Further moderator analyses (see Table 6) identified several contextual factors influencing intervention effectiveness. Significantly larger effects were found for interventions conducted with secondary school students (g = 0.636), in natural science disciplines (g = 0.753), within fully online environments (g = 0.732), and for durations of less than two weeks (g = 0.776).

Discussion and Significance

This meta-analysis moves the field beyond the question of whether artificial intelligence supports SRL to provide a nuanced, evidence-based account of how, when, and which types of AI are most effective. By synthesizing a decade of research through the dual lenses of Zimmerman's (2002) SRL model and AI's technological evolution, this study offers significant theoretical and practical contributions.

Bridging the Performance-Competence Divide: A Core Challenge

The central finding of this study is the identification of a critical "performance-competence divide." While our results confirm that AI interventions yield a moderate overall benefit (g = 0.507), their impact is strongest during the in-task performance phase. This suggests that AI excels at providing immediate, in-the-moment scaffolding. However, our synthesis of behavioral evidence indicates that these supported regulatory behaviors often decay once the technology is withdrawn (e.g., Bouchet et al., 2016; Long & Aleven, 2013). This pattern echoes Salomon et al.'s (1991) seminal distinction between the effects with technology versus the lasting cognitive residue, or effects of technology. Our findings empirically substantiate this divide in the context of modern AI, suggesting current systems are more adept at acting as cognitive partners than as tools for internalizing durable skills.

Furthermore, this study demonstrates that AI is not a monolithic entity. The superior efficacy of generative AI, especially in the historically challenging forethought (g = 0.709) and self-reflection (g = 0.826) phases, signals a paradigm shift. Its effectiveness stems from its capacity for dialogic interaction, which can scaffold the nuanced processes of goal-setting and causal attribution in ways that passive data dashboards cannot (Lee et al., 2024; Pan et al., 2025). This underscores that AI's value lies not in its raw power, but in the precise alignment between its affordances and the cognitive demands of each SRL phase (see Table 5).

Implications for Design: Engineering AI to Cultivate Autonomy

These insights carry profound implications for designing the next generation of educational technologies. To bridge the performance-competence divide, the field must move beyond creating systems that risk fostering dependency and instead engineer AI that explicitly cultivates learner autonomy. Our findings advocate for intelligent integration strategies, such as combining the stable, structural scaffolding of rule-based systems with the adaptive, dialogic support of generative AI to address the full SRL cycle. Critically, AI systems must be designed with mechanisms that promote skill internalization. This includes implementing adaptive scaffolds that strategically fade over time, utilizing guidance-based approaches that provide hints rather than direct answers (Lee et al., 2025), and creating opportunities for "productive failure" where support is temporarily withdrawn to encourage autonomous problem-solving.

A Research Agenda: Moving Beyond Immediate Performance

Finally, our work calls for a fundamental shift in how the field researches and assesses AI-supported SRL. The moderator analysis revealing the outsized impact of short-term interventions (<2 weeks, g = 0.776) points to a prevalence of novelty effects that may mask the development of lasting skills. To generate more valid insights, future research must adopt longitudinal designs to track whether scaffolded behaviors persist after the intervention ends. This necessitates new assessment frameworks that prioritize delayed, unsupported performance as the ultimate indicator of success. The critical question for the field must evolve from "Does AI improve learning outcomes?" to "Does AI cultivate learners who can regulate their own learning independently?". Only by addressing this question can we harness AI's potential to develop the authentic, transferable autonomy that underpins lifelong learning.

Figures and Tables

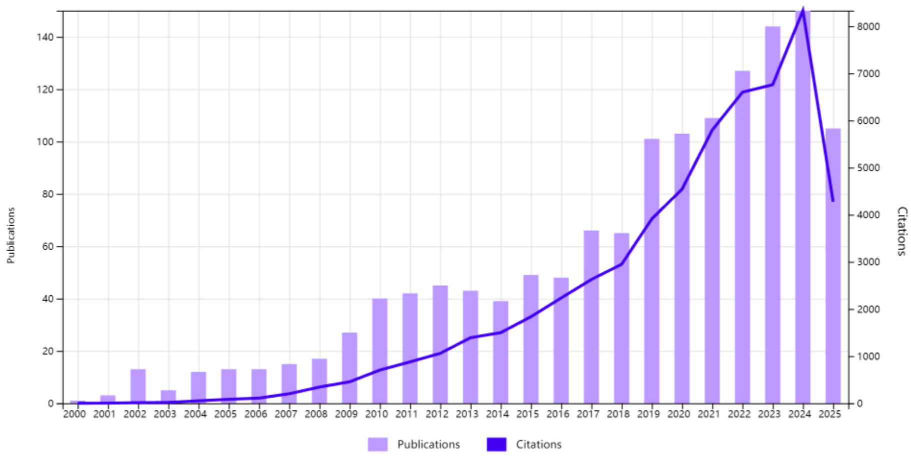

Figure 1. Annual Publications and Citation Trends in Self-Regulated Learning Since the 21st Century

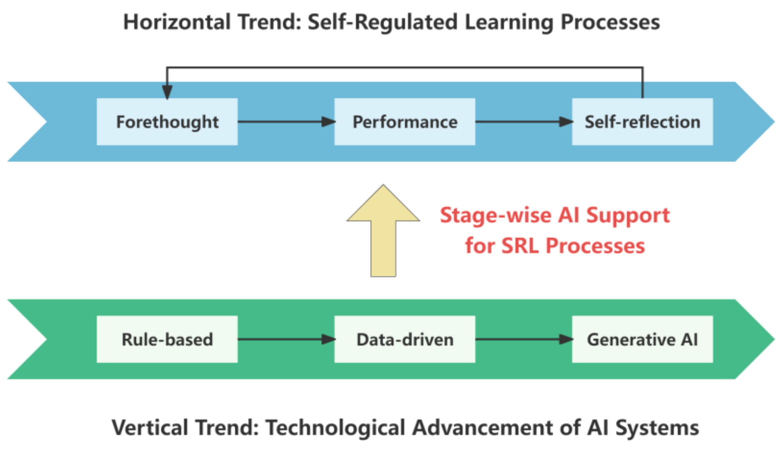

Figure 2. The Two Key Dimensions of This Study

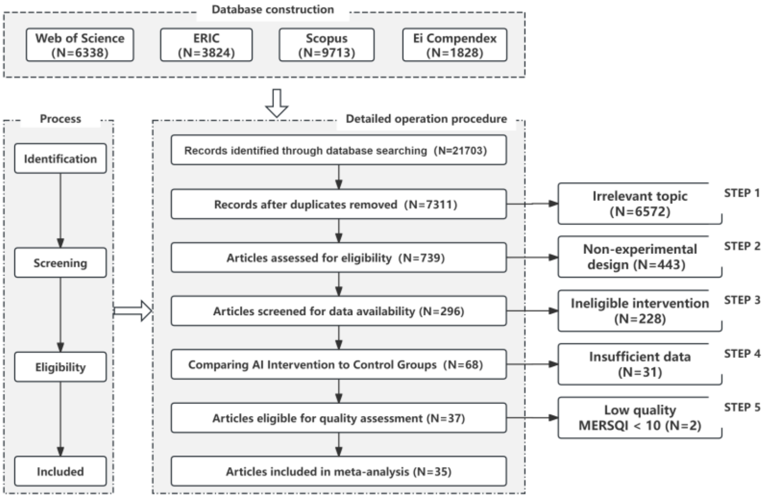

Figure 3. Search flow diagram

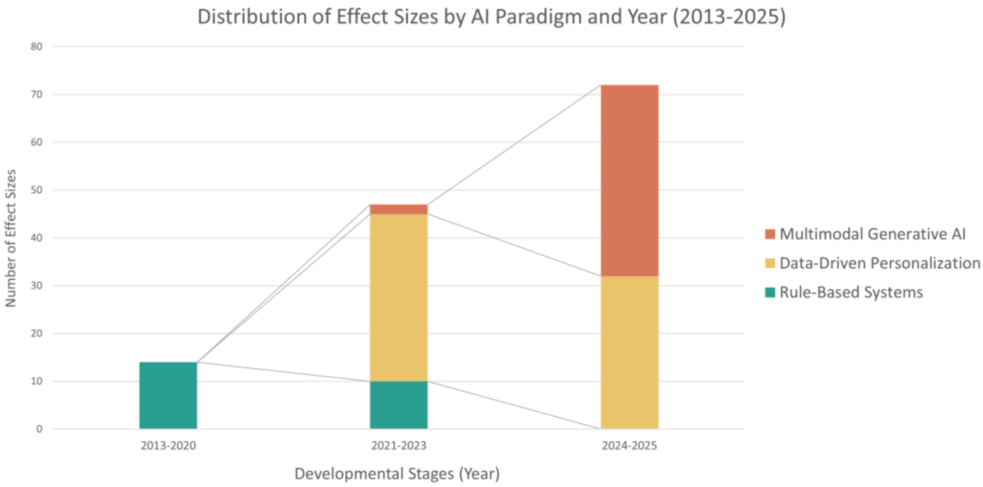

Figure 4. Distribution of Effect Sizes by AI Paradigm and Year (k=133 effect sizes)

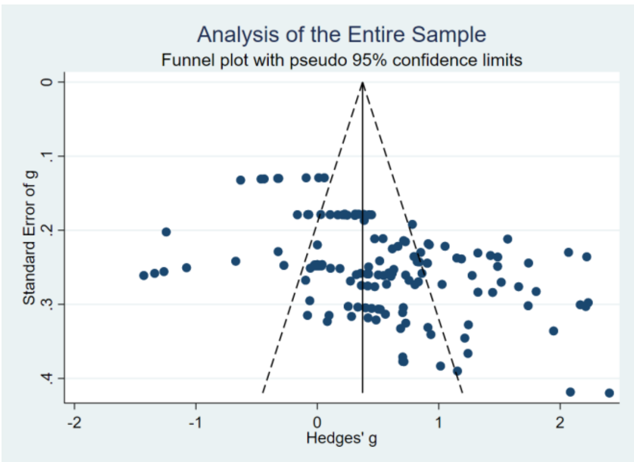

Figure 5. Funnel Plot of Overall Sample

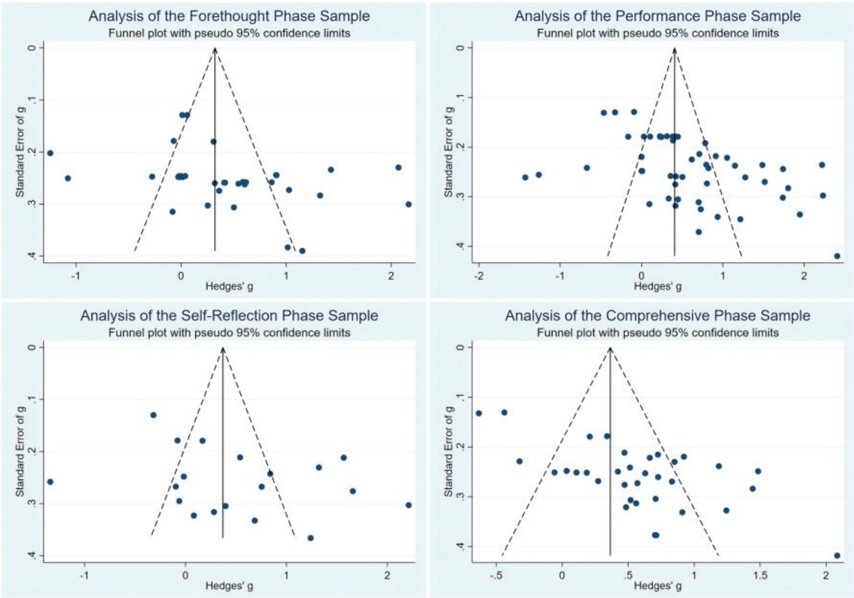

Figure 6. Funnel Plot of Combined Phases

Table 1. Lists of codes for the analysis of selected articles

Dimension	Item	Description
Participant Characteristics	Region	Text records, e.g., the United States, China, Japan, etc.
	Sample Size	(1) 1–50; (2) 51–100; (3) 101–300.
	Educational Level	(1) Primary school; (2) Junior and senior high school; (3) Institutions of higher education.
	Discipline	(1) Natural Science (including science, mathematics, physics, biology, geography); (2) Social Science (including politics, education, psychology, linguistics).
Content Characteristics	Learning Environment	(1) Online learning settings; (2) Blended learning settings.
	AI Intervention Method	A system that uses computational models (e.g., rule-based logic, machine learning, Generative Modeling) to provide adaptive or personalized feedback and scaffolding for SRL.
	SRL Variables	Variables related to SRL recorded in the literature, such as self-efficacy, metacognition, attention focus, and time management.

Table 2. Mapping of SRL Indicators to Phases

Phase	Theoretical Definition	Indicator	Definition
Forethought (Pre-task)	The preparatory phase involving task analysis and mobilizing motivational beliefs.	Goal setting and strategy planning	Establishing clear learning goals and implementation paths.
		Self-efficacy	Confidence in one's ability to complete tasks.
		Task value and interest	Awareness of the significance and enjoyment of learning tasks.
		Outcome expectation	Predictions of the effort and potential benefits of learning.
Performance (During task)	The execution phase involving self-control (e.g., strategy use) and self-observation.	Task strategies	Applying cognitive and metacognitive skills (e.g., summarizing, monitoring).
		Attention focus and time management	Focusing on tasks and allocating study time effectively.
		Self-observation and monitoring	Continuously tracking one's performance and strategy usage.
		Seeking help	Identifying difficulties and actively seeking external assistance.
Self-Reflection (Post-task)	The evaluative phase involving self-judgment of performance and adaptive self-reactions.	Self-evaluation	Comparing outcomes with standards or initial expectations.
		Causal attribution	Interpreting successes or failures as a basis for strategy modification or effort adjustment.
		Adaptive inference and	Adjusting future study behavior based on

		response	reflective results.
Comprehensive Phase	An assessment of the overall effectiveness of an SRL-driven task, using measures not attributable to a single phase.	Academic achievement	Test results or indicators of academic performance.
		Skill acquisition and transfer	Application of skills and knowledge in new contexts.
		Learner satisfaction and engagement	Emotional responses and investment in the learning process.

Table 3. Random-Effects Model for AI Interventions in SRL Phases

	Effect size and 95% CI					Heterogeneity test				Tau-squared		
Variable	*k*	*Hedges' g*	*SE*	*Lower*	*Upper*	*Q*	*df(Q)*	*p*	*I²*	*Tau-squared*	*SE*	*Tau*
Forethought Phase	33	0.401	0.123	0.160	0.643	271.43	32	0.0000	88.95	0.415	0.133	0.644
Performance Phase	51	0.574	0.110	0.358	0.789	533.81	50	0.0000	92.05	0.474	0.119	0.688
Self-Reflection Phase	18	0.464	0.196	0.079	0.848	194.45	17	0.0000	91.81	0.594	0.256	0.771
Comprehensive Phase	31	0.529	0.103	0.328	0.730	204.14	30	0.0000	81.93	0.337	0.117	0.581
Overall	133	0.507	0.063	0.384	0.630	1208.24	132	0.0000	89.81	0.432	0.069	0.657

Table 4. Measures of publication bias

		Egger's				Fail-Safe N	
Self-Regulated Learning Process	*k*	Score	*SD*	*z*	*p*	Classic	Orwin
Forethought Phase	33	5.15	2.270	2.27	0.0232	594	930
Performance Phase	51	5.94	1.622	3.66	0.0003	3167	1945
Self-Reflection Phase	18	4.08	3.333	1.22	0.2213	218	555
Comprehensive Phase	31	5.86	3.384	1.73	0.0834	843	930
Overall	133	4.66	1.186	3.93	0.0001	5908	4451

[a] k, number of studies

[b] **Classic Fail-Safe N:** Represents the number of null-effect studies needed to raise the p-value above the significance threshold ($\alpha = 0.05$).

[c] **Orwin's Fail-Safe N:** Represents the number of missing studies required to reduce the effect size to a predefined trivial level (ES = 0.01).

Table 5. Combined Effect Sizes for AI Developmental Stages and SRL Phases (Hedges' g)

SRL Process	AI Phase	k	Hedges' g	95% CI	I²	p	Z
Forethought	Rule-based	2	-	-	-	-	-
	Data-driven	15	0.083	[-0.301, 0.466]	92.53 %	0.6708	0.43
	Generative	16	0.709	[0.437, 0.981]	76.84 %	0.0000	5.11
Performance	Rule-based	6	-	-	-	-	-
	Data-driven	32	0.352	[0.080, 0.624]	93.45 %	0.0112	2.54
	Generative	13	0.938	[0.549, 1.328]	85.64 %	0.0000	4.71
Self-Reflection	Rule-based	2	-	-	-	-	-
	Data-driven	8	0.196	[-0.387, 0.779]	94.24 %	0.5105	0.66
	Generative	8	0.826	[0.247, 1.405]	88.08 %	0.0051	2.80
Comprehensive	Rule-based	13	0.429	[0.197, 0.662]	59.66 %	0.0003	3.61
	Data-driven	13	0.580	[0.190, 0.971]	90.94 %	0.0036	2.91
	Generative	8	0.682	[0.416, 0.938]	60.22%	0.0000	5.02
Overall	Rule-based	23	0.534	[0.339, 0.730]	66.77 %	0.0000	5.34
	Data-driven	68	0.318	[0.134, 0.501]	93.07 %	0.0006	3.41
	Generative	42	0.799	[0.604, 0.994]	82.34 %	0.0000	8.03

[a] Caution is advised for subgroups with $k < 10$, while those with $k < 7$ may have limited reliability.

Table 6. Subgroup Analysis Results of Meta-Analysis(Hedges' g)

Subgroup	**k**	**g**	**95 % CI**	**Z (p)**	**Between p***
Educational level					**0.221 (3.02)**
Primary school	14	0.304	[0.060 , 0.548]	2.44 (0.015)	
Junior and senior high school	11	0.636	[0.284 , 0.989]	3.54 (0.001)	
Higher education	108	0.517	[0.374 , 0.660]	7.09 (< 0.001)	
Subject					**< 0.001 (21.09)**
Natural Science	73	0.753	[0.576 , 0.929]	8.36 (< 0.001)	
Social Science	56	0.211	[0.061 , 0.360]	2.77 (0.006)	
Combined subject areas	129	0.516	[0.390 , 0.643]	8.00 (< 0.001)	
Learning space					**0.087 (2.93)**
Online learning	37	0.732	[0.376 , 1.087]	4.04 (< 0.001)	
Blended learning	96	0.409	[0.309 , 0.509]	8.02 (< 0.001)	
Experimental duration					**0.002 (12.44)**
Less than 2 weeks	45	0.776	[0.551 , 1.002]	6.88 (< 0.001)	
2 – 10 weeks	43	0.361	[0.222 , 0.500]	7.64 (< 0.001)	
> 10 weeks	45	0.374	[0.139 , 0.609]	0.32 (< 0.001)	
Overall	**133**	**0.507**	**[0.384 , 0.630]**	**8.08 (< 0.001)**	—

Funding

This study is supported by a 2021 major project from the National Social Science Foundation of China (grant number 21 and ZD328).

Acknowledgments

This research has been funded by the Research Council of Finland (formerly Academy of Finland) grant numbers 324381, 350249, the University of Oulu profiling project Profi7 Hybrid Intelligence - 352788. This work/Part of the work was carried out with the support of LeaF Research Infrastructure, University of Oulu, Finland.

References

Bandura, A. (1991). Social cognitive theory of self-regulation. *Organizational Behavior and Human Decision Processes*, *50*(2), 248–287. https://doi.org/10.1016/0749-5978(91)90022-L

Bouchet, F., Harley, J. M., & Azevedo, R. (2016). Can Adaptive Pedagogical Agents' Prompting Strategies Improve Students' Learning and Self-Regulation? In A. Micarelli, J. Stamper, & K. Panourgia (Eds.), *Intelligent Tutoring Systems* (pp. 368–374). Springer International Publishing. https://doi.org/10.1007/978-3-319-39583-8_43

Chen, S., Cheng, H., & Huang, Y. (2024). Emotion Recognition in Self-Regulated Learning: Advancing Metacognition Through AI-Assisted Reflections. In D. Kourkoulou, A.-O. (Olnancy) Tzirides, B. Cope, & M. Kalantzis (Eds.), *Trust and Inclusion in AI-Mediated Education: Where Human Learning Meets Learning Machines* (pp. 185–212). Springer Nature Switzerland. https://doi.org/10.1007/978-3-031-64487-0_9

Dent, A. L., & Koenka, A. C. (2016). The Relation Between Self-Regulated Learning and Academic Achievement Across Childhood and Adolescence: A Meta-Analysis. *Educational Psychology Review*, *28*(3), 425–474. Scopus. https://doi.org/10.1007/s10648-015-9320-8

Dignath, C., & Büttner, G. (2018). Teachers' direct and indirect promotion of self-regulated learning in primary and secondary school mathematics classes – insights from video-based classroom observations and teacher interviews. *Metacognition and Learning*, *13*(2), 127–157. Scopus. https://doi.org/10.1007/s11409-018-9181-x

Duffy, M. C., & Azevedo, R. (2015). Motivation matters: Interactions between achievement goals and agent scaffolding for self-regulated learning within an intelligent tutoring system. *Computers in Human Behavior*, *52*, 338–348. https://doi.org/10.1016/j.chb.2015.05.041

Harati, H., Sujo-Montes, L., Tu, C.-H., Armfield, S. J. W., & Yen, C.-J. (2021). Assessment and Learning in Knowledge Spaces (ALEKS) Adaptive System Impact on Students' Perception and Self-Regulated Learning Skills. *Education Sciences*, *11*(10), Article 10. https://doi.org/10.3390/educsci11100603

Hedges, L. V., & Tipton, E. (2010). Meta-analysis. In A. Steptoe (Ed.), *Handbook of Behavioral Medicine* (pp. 909–921). Springer New York. https://doi.org/10.1007/978-0-387-09488-5_58

Higgins, J. P. T., & Thompson, S. G. (2002). Quantifying heterogeneity in a meta-analysis. *Statistics in Medicine*, *21*(11), 1539–1558. https://doi.org/10.1002/sim.1186

Landis, J. R., & Koch, G. G. (1977). The measurement of observer agreement for categorical data. *Biometrics*, 159–174.

Lee, H.-Y., Chen, P.-H., Wang, W.-S., Huang, Y.-M., & Wu, T.-T. (2024). Empowering ChatGPT with guidance mechanism in blended learning: Effect of self-regulated learning, higher-order thinking skills, and knowledge construction. *International Journal of Educational Technology in Higher Education*, *21*(1), 16. https://doi.org/10.1186/s41239-024-00447-4

Lee, Y.-F., Hwang, G.-J., & Chen, P.-Y. (2025). Technology-based interactive guidance to promote learning performance and self-regulation: A chatbot-assisted self-regulated learning approach. *Educational Technology Research and Development*. https://doi.org/10.1007/s11423-025-10478-x

Long, Y., & Aleven, V. (2013). Supporting Students' Self-Regulated Learning with an Open Learner Model in a Linear Equation Tutor. In H. C. Lane, K. Yacef, J. Mostow, & P. Pavlik (Eds.), *Artificial Intelligence in Education* (pp. 219–228). Springer. https://doi.org/10.1007/978-3-642-39112-5_23

Lu, W., & Lin, C. (2024). Meta-Analysis of Influencing Factors on the Use of Artificial Intelligence in Education. *The Asia-Pacific Education Researcher*. https://doi.org/10.1007/s40299-024-00883-w

Ng, D. T. K., Tan, C. W., & Leung, J. K. L. (2024). Empowering student self-regulated learning and science education through CHATGPT: A pioneering pilot study. *British Journal of Educational Technology*, *55*(4), 1328–1353. https://doi.org/10.1111/bjet.13454

Pan, M., Lai, C., & Guo, K. (2025). Effects of GenAI-empowered interactive support on university EFL students' self-regulated strategy use and engagement in reading. *The Internet and Higher Education*, *65*, 100991.

Pigott, T. D., & Polanin, J. R. (2020). Methodological Guidance Paper: High-Quality Meta-Analysis in a Systematic Review. *Review of Educational Research*, *90*(1), 24–46. https://doi.org/10.3102/0034654319877153

SALOMON, G., PERKINS, D. N., & GLOBERSON, T. (1991). Partners in Cognition: Extending Human Intelligence with Intelligent Technologies. *Educational Researcher*, *20*(3), 2–9. https://doi.org/10.3102/0013189X020003002

Sawatsky, A. P., Beckman, T. J., Edakkanambeth Varayil, J., Mandrekar, J. N., Reed, D. A., & Wang, A. T. (2015). Association Between Study Quality and Publication Rates of Medical Education Abstracts Presented at the Society of General Internal Medicine Annual Meeting. *Journal of General Internal Medicine*, *30*(8), 1172–1177. https://doi.org/10.1007/s11606-015-3269-7

Winne, P., & Hadwin, A. (1998). Studying as Self-Regulated Learning. In *Metacognition in Educational Theory and Practice* (Vol. 93, pp. 277–304).

Wong, J., Baars, M., Davis, D., Zee, T. V. D., Houben, G.-J., & Paas, F. (2019). Supporting Self-Regulated Learning in Online Learning Environments and MOOCs: A Systematic Review. *International Journal of Human–Computer Interaction*. https://www.tandfonline.com/doi/abs/10.1080/10447318.2018.1543084

Zimmerman, B. J. (1989). A social cognitive view of self-regulated academic learning. *Journal of Educational Psychology*, *81*(3), 329–339. https://doi.org/10.1037/0022-0663.81.3.329

Zimmerman, B. J. (2002). Becoming a Self-Regulated Learner: An Overview. *Theory Into Practice*, *41*(2), 64–70. https://doi.org/10.1207/s15430421tip4102_2